Palatial Pools Energy Comparison

Neptune & Echo vs Leading Competitors (2025)

Heat Pump Comparison (13-14kW Range)

Based on 50,000L pool, 8 months usage, 6 hours/day, 15°C average ambient temp, $0.33/kWh.

Brand & Model	Compressor Tech	COP @ 15°C	Est. Annual kWh	Est. Annual Cost	vs Neptune X-Pro
Neptune X-Pro (13.2kW)	Mitsubishi DC Inverter	8.5	2,263 kWh	$746	Benchmark
Neptune X-Eco (12.5kW)	Mitsubishi DC Inverter	7.0	2,598 kWh	$857	+$111 / yr
Madimack InverELITE V4 (14.2kW)	Twin Rotary Mitsubishi	8.5	2,438 kWh	$804	+$58 / yr
Pentair UltraTemp HXi (16kW)	Mitsubishi/Panasonic	5.8	3,171 kWh	$1,046	+$300 / yr
AstralPool ECO Inverter (12kW)	Standard Inverter	5.1	3,120 kWh	$1,029	+$283 / yr
Hayward HeatPro (14.4kW)	Scroll Compressor	4.5	4,672 kWh	$1,541	+$795 / yr

Pool Pump Comparison (1.5HP Range)

Based on 50,000L pool, 8 hours/day filtration, $0.33/kWh.

Brand & Model	Energy Rating	Typical Wattage	Est. Annual kWh	Est. Annual Cost	vs Neptune X-Eco
Neptune X-Eco (1.5HP)	6-Star	180 W	416 kWh	$137	Benchmark
Neptune X-Pro (1.5HP)	6-Star	160 W	390 kWh	$128	Save $9 / yr
Pentair IntelliFlo3 VSF (1.5HP)	WEF 9.4	220 W	480 kWh	$158	+$21 / yr
Hayward TriStar VS (1.5HP)	6-Star	250 W	547 kWh	$180	+$43 / yr
AstralPool CTX 400C (1.5HP)	1-Star	1,650 W	1,572 kWh	$518	+$381 / yr
Generic Single Speed (1.5HP)	1-Star	1,500 W	1,800 kWh	$594	+$457 / yr

Methodology: Running costs calculated using Melbourne 2025 residential default offer rate (~$0.33/kWh). Heat pump COP values sourced from manufacturer datasheets at 15°C ambient. Pool pump energy consumption based on Australian Government Energy Rating label data. Actual savings will vary based on pool insulation, climate, and usage patterns.